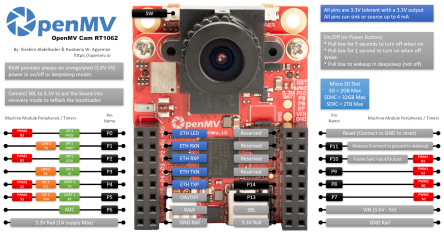
OpenMV® Cam RT1062

Datasheet

SKU: OMV-CAM-RT1062-V1/V2/V3 (PCB V4.2/V5/V6)

Description

The OpenMV Cam RT1062 is built around NXP's high-performance i.MX RT1062 crossover MCU running at 600 MHz, combining the real-time capabilities of a microcontroller with application processor-level performance. It features 32 MB external SDRAM, 16 MB QUADSPI flash, the OV5640 5 MP autofocus camera, integrated WiFi 802.11 a/b/g/n and Bluetooth 5.1, 10/100 Ethernet, a secure element, accelerometer, and USB-C High-Speed connectivity. Programmable in MicroPython via the OpenMV IDE with TensorFlow Lite support.



Target Areas

Machine vision, edge AI, object detection, thermal camera, video streaming, industrial automation, robotics, IoT, security, barcode/QR scanning.

Key Features

- **600 MHz crossover MCU** — NXP i.MX RT1062 ARM Cortex-M7 at 600 MHz with hardware FPU and DSP.
- **Extended memory** — 32 MB SDRAM (320 MB/s) and 16 MB QUADSPI flash (66 MB/s) for large frame buffers and AI models.
- **5 MP camera** — OV5640 autofocus camera module. M12 interchangeable lens mount. Removable module system.
- **WiFi + Bluetooth** — 802.11 a/b/g/n WiFi with chip antenna (U.FL alternative). Bluetooth 5.1 (BR/EDR and BLE).
- **Ethernet** — 10/100 Mbps Ethernet with PoE support option.
- **Security** — NXP SE050 secure element for cryptographic protection and authentication.
- **USB-C High-Speed** — USB 2.0 High-Speed (480 Mbps) with EMI filtering and TVS protection. 1.5 A current limited.
- **Ultra-low power** — Less than 30 µA deepsleep current with on-board RTC. Power button on/off support.

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1 Features and System Components

1.1 Microcontroller

Component	Details
i.MX RT1062	NXP i.MX RT1062 crossover MCU with ARM Cortex-M7 core running at 600 MHz. Features hardware floating-point unit (FPU) and DSP instructions. Includes 1 MB on-chip SRAM.

1.2 Memory

Component	Details
On-chip SRAM	1 MB on-chip SRAM.
External SDRAM	32 MB SDRAM (IS42S16160G), 16-bit bus at 160 MHz (320 MB/s bandwidth).
External QUADSPI Flash	16 MB QUADSPI NOR flash (IS25LPWP128F), 133 MHz (66 MB/s read).
microSD Card	Supports SD (up to 2 GB), SDHC (up to 32 GB), SDXC (up to 2 TB). 25 MB/s read/write.

1.3 Connectivity

Component	Details
USB-C	USB 2.0 High-Speed (480 Mbps). 1.5 A current limited. EMI filtering (ECMF02-4CMX8) and TVS protection.
WiFi	802.11 a/b/g/n via CYW4334 combo chip. On-board chip antenna with U.FL connector as alternative.
Bluetooth	Bluetooth 5.1 (BR/EDR and BLE) via same WiFi/BT combo chip.
Ethernet	10/100 Mbps via KSZ8081RND PHY. PoE support option. Differential pairs (TXP/TXN, RXP/RXN) on header.
SPI	SPI bus on P0 (SS) / P1 (MOSI) / P2 (MISO) / P3 (SCLK). Up to 60 Mb/s.
I2C	I2C bus on P4 (SCL) / P5 (SDA). Up to 1 Mb/s.
CAN	CAN bus on P4 (TX) / P5 (RX). Up to 1 Mb/s.
UART	UART on header pins. Up to 20 Mb/s.

1.4 Camera System

Component	Details
Image Sensor	OV5640 1/4-inch 5 MP CMOS image sensor (2592x1944) with autofocus.
Image Performance	~40 FPS at QVGA (320x240) for simple algorithms. Full 5 MP JPEG capture.
Lens	M12 (S-mount) interchangeable lens mount.
Camera Interface	Removable camera module system. Compatible with alternative sensor modules.

1.5 Security

Component	Details
SE050C1HQ1	NXP SE050 secure element for cryptographic protection, secure authentication, and key storage.

1.6 Sensors

Component	Details
Accelerometer	MXC4005XC 3-axis accelerometer. 12-bit resolution, 2/4/8g selectable range.

1.7 LEDs

Component	Details
User RGB LED	1 user-controllable and dimmable RGB LED.
Power/Charge RGB LED	1 separate RGB LED for charge and power status indication.

1.8 Power Management

Component	Details
USB-C	Power input via USB-C. 1.5 A current limited.
VIN	External power input, 3.6 V to 5.0 V. Reverse supply protection via LM66200 ideal diode.
RAW	Always-on unregulated 3.6–5 V power. Active in on, off, and deepsleep modes.
LiPo Battery	3.7 V single-cell LiPo connector. 100 mA fast charge via BQ24075 charger IC.
Voltage Regulator	AP61300 step-down converter plus TCR2LE31 LDO.
Low-Power Mode	Less than 30 µA deepsleep current. On-board RTC. Power button: hold 5s to off, 1s to on.

1.9 Debug

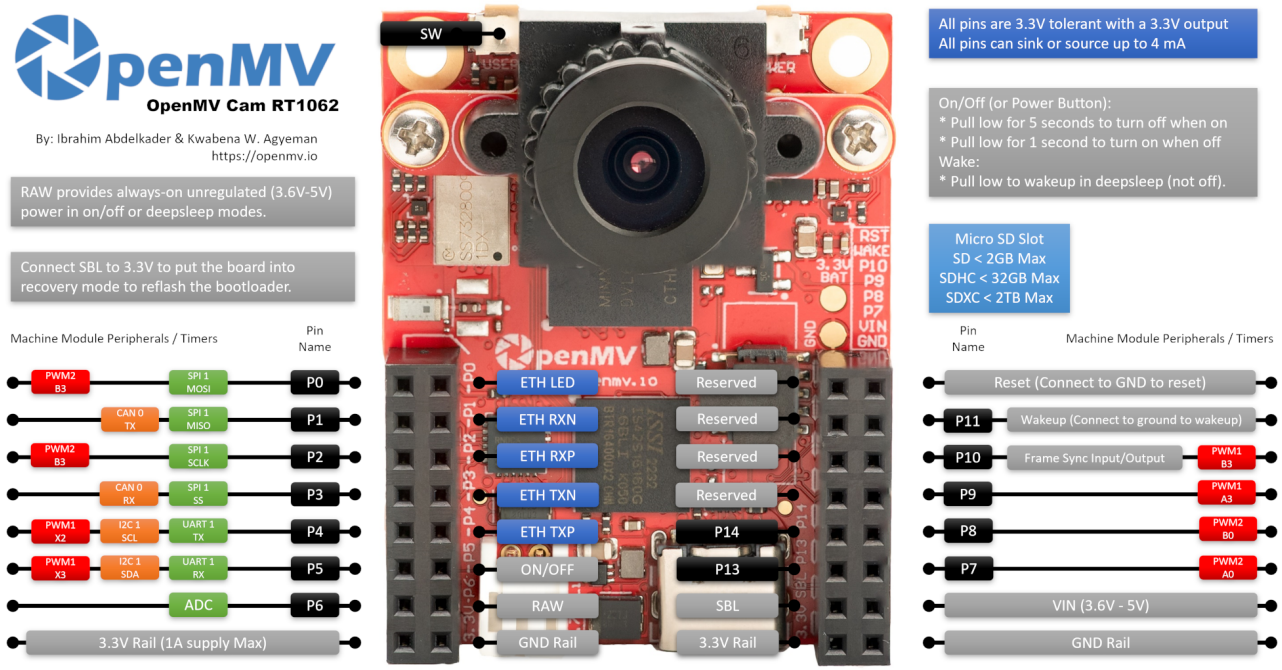
Component	Details
JTAG	ARM 10-pin JTAG header. SEGGER J-Link compatible.

2 Ratings

2.1 Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Unit
VUSB	USB-C supply voltage	4.5	5.0	5.5	V
VIN	VIN supply voltage	3.6	5.0	5.0	V
VIO	I/O voltage	—	3.3	3.3	V
IIO	Per-pin current (sink/source)	—	—	4	mA
IOUT	3.3 V rail max current	—	—	1000	mA
ISLEEP	Deepsleep current	—	—	30	µA
TOP	Operating temperature	-40	25	105	°C

3 Connector Overview



3.1 Left Header (J1)

Pin	Label	Type	Alternate Functions
1	P0	I/O	SPI SS
2	P1	I/O	SPI MOSI
3	P2	I/O	SPI MISO
4	P3	I/O	SPI SCLK
5	P4	I/O	I2C SCL / CAN TX
6	P5	I/O	I2C SDA / CAN RX
7	P6	I/O	ADC (12-bit, 3.3V tolerant)
8	3V3	Power	3.3 V regulated output (1 A max)
9	GND	Power	Ground

3.2 Right Header (J2)

Pin	Label	Type	Alternate Functions
1	P11	Input	Reset (connect to GND to reset)
2	P10	I/O	Wakeup (connect or ground to wake)
3	P9	I/O	Frame Sync (input/output)
4	P8	I/O	Servo / PWM
5	P7	I/O	Servo / PWM
6	P14	I/O	UART TX (secondary)
7	VIN	Power	3.6–5.0 V input
8	GND	Power	Ground

3.3 Ethernet Header Pins

Pin	Label	Type	Function
1	ETH-TXP	Ethernet	TX positive
2	ETH-TXN	Ethernet	TX negative
3	ETH-RXP	Ethernet	RX positive
4	ETH-RXN	Ethernet	RX negative
5	ETH-LED	Output	Ethernet status LED

4 Functional Overview

4.1 Power Tree

The OpenMV Cam RT1062 can be powered from USB-C (1.5 A limited), VIN (3.6\u20135.0 V with LM66200 reverse protection), RAW (always-on, active in all power modes), or a 3.7 V LiPo battery (BQ24075 charger, 100 mA fast charge). The AP61300 step-down converter and TCR2LE31 LDO provide regulated 3.3 V. Deepsleep current is less than 30 \u00B5A with on-board RTC. Power button supports hold-to-off (5s) and press-to-wake (1s).

5 General Info

Dimension	Value
Weight	70 g
Lens Mount	M12 (S-mount), interchangeable
Mounting	2 mounting holes
Operating Temperature	-40°C to +85°C

6 Certifications

Certification	Region	Details
RoHS	EU	Compliant with EU Directive 2011/65/EU (RoHS 2) and amendment 2015/863 (RoHS 3). All components and solder are lead-free.
REACH	EU	Compliant with EU Regulation (EC) No 1907/2006.

7 Company Information

Field	Details
Website	https://openmv.io
Documentation	https://docs.openmv.io
Support	https://forums.openmv.io

8 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/openmv-cam-rt
Documentation	https://docs.openmv.io
Community Forum	https://forums.openmv.io
i.MX RT1062	https://www.nxp.com/products/processors-and-microcontrollers/arm-microcontrollers/i-mx-rt-crossover-mcus/i-mx-rt1060-crossover-mcu:i.MX-RT1060
OV5640 Image Sensor	https://www.ovt.com/products/sensor.php?id=101
OpenMV IDE	https://openmv.io/pages/download

9 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.